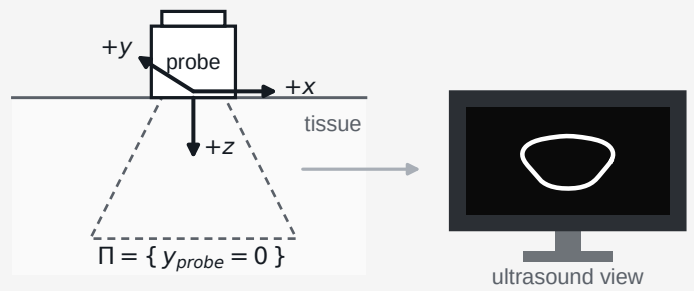


(a) Probe frame / image plane



Probe-frame twist

$$V_{probe} = (v_x, v_y, v_z, \omega_x, \omega_y, \omega_z)$$

$$\text{Stab}(\Pi) = \text{span}\{v_x, v_z, \omega_y\}$$

$$\text{Off}(\Pi) = \text{span}\{v_y, \omega_x, \omega_z\}$$

Motion classes

In-plane: same section

Out-of-plane: new section

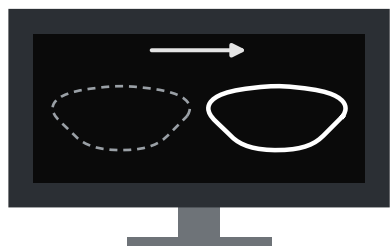
Metric depth

(b) In-plane

$$\text{Stab}(\Pi) = \text{span}\{v_x, v_z, \omega_y\}$$

$$v_x$$

Lateral



$$\dot{x}_g = -v_x - \omega_y z_g$$

Image policy

Analytic

$$v_z$$

Normal press



$$\dot{z}_g = -v_z + \omega_y x_g$$

Admittance

Force control

$$\omega_y$$

In-plane rotation



$$\dot{\theta} = \omega_y$$

Analytic

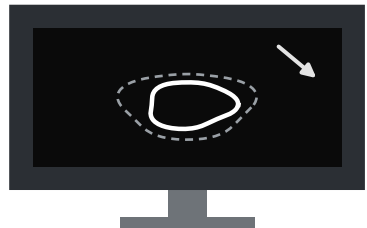
$$M_y \rightarrow 0$$

(c) Out-of-plane

$$\text{Off}(\Pi) = \text{span}\{v_y, \omega_x, \omega_z\}$$

$$v_y$$

Elevation



New section

Image policy

No 1st-order gain

$$\omega_x$$

Tilt



Plane tilt

Admittance

$$M_x \rightarrow 0$$

$$\omega_z$$

Spin



Beam-axis spin

Image policy

Axis $\parallel v_y$

(d) Axis allocation

$$V_{probe} = [v_x^{img}, v_y^{img}, v_z^{adm}, \omega_x^{adm}, \omega_y^{adm}, \omega_z^{img}]$$

Image: ~5 Hz (ZOH) · Force: 100 Hz